Jared Macshane

PhD Student in Computer Science

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Education

Current **PhD Computer Science**, *University of California, Irvine*
- Distributed Systems Middleware Group

Master of Computer Science, *California State University San Marcos*
- Thesis: Using Growing Self-organizing Maps to Construct Trail Networks Using Public GPS Data

Bachelor of Mathematics, *University of California, Santa Barbara*

Professional Experience

2022 **Machine Learning Consultant**, *Ecorithms*
- Architected and deployed production-ready semantic segmentation models for aerial imagery
- Optimized inference pipeline through model compression and deployment strategies

2021 **Research Fellow**, *San Diego Zoo Wildlife Alliance - Conservation Lab*
- Developed and deployed real-time computer vision system for wildlife monitoring at scale
- Engineered multi-threaded pipeline for processing high-volume image streams
- Automated manual review process through intelligent alert system

2018–2020 **Ad Account Manager**, *The New Patient Machine*
- Managed digital marketing campaigns and client relationships for healthcare practices
- Analyzed campaign performance metrics to optimize client ROI
- Developed and implemented strategic marketing solutions across multiple platforms

2017–2018 **Mathematics Tutor**, *Mathnasium*
- Provided individualized instruction in mathematics from elementary through calculus level
- Developed customized learning plans to address specific student needs
- Maintained detailed progress reports and communicated effectively with parents and staff

Awards & Achievements

2024 Dean's Fellowship - Competitive full-funding award for outstanding PhD candidates

2024 NSF Advanced Studies Institute (ASI) Fellowship - Selected for intensive coastal zone hazards research program at Tohoku University, Japan

2023 Best Paper Award - IEEE SIEDS for novel machine learning approach to geospatial data

Research Projects

2024–Present **SHIELD Digital Twin Project**, *University of California, Irvine*

- Leading development of large-scale digital twin system for communities
- Architecting distributed edge computing framework for efficient network utilization
- Implementing novel progressive inference models for resource-constrained environments
- Coordinating cross-functional team across multiple institutions

2023–2024 **Wildfire Intelligence Projects**, *University of California, Irvine*

- Developed generative model for wildfire spread prediction using Conditional Variational Autoencoders
- Designed supervised compression system achieving 72.9% detection accuracy while reducing image size to 4.8KB
- Designed edge computing framework for distributed wildfire detection using early-exit neural networks

2021–2023 **Hands-On Virtual Reality Labs**, *California State University San Marcos*

- Developed computer vision tracking system using Aruco markers for precise spatial tracking
- Led user experience studies to validate system effectiveness and improve user interaction

2022 **Geospatial Machine Learning**, *California State University San Marcos*

- Designed novel machine learning algorithm for processing large-scale geospatial data
- Developed automated trail detection system for noisy GPS traces
- Implemented efficient data structures for improved computation performance

Publications

In Preparation Macshane, J., et al. "SHIELD: A Distributed Edge Computing Framework for Community-Scale Digital Twins." To be submitted

Jiang, Y., Hamadani, K., Ng, K., Ahmadinia, A., Aquino, A., Palacio, R., Huang, J., Macshane, J., & Hadaegh, A. "Developing scalable hands-on virtual and mixed-reality science labs." Virtual Reality 28, 173 (2024).

Gow, Sean, et al. "Miniaturization and geometric optimization of SteamVR active optical trackers." Optical Architectures for Displays and Sensing in Augmented, Virtual, and Mixed Reality (AR, VR, MR) IV. Vol. 12449. SPIE, 2023.

Macshane, Jared, and Ali Ahmadinia. "AI Assisted Trail Map Generation based on Public GPS Data." 2023 Systems and Information Engineering Design Symposium (SIEDS). IEEE, 2023.

Academic and Mentoring Experience

2023–Current **Teaching Assistant**, *University of California, Irvine*

- ICS 32: Programming with Software Libraries (F23,W25)
- ICS 33: Intermediate Programming (F24)
- CSE 90: Systems Engineering and Technical Communications (W24)

2023–Present **Graduate Student Mentor**, *Zotbins Undergraduate Research Project*, University of California, Irvine

- Leading team of undergraduate researchers developing IoT-enabled smart waste management system
- Managing project lifecycle from design through iterative deployments across campus locations
- Developing ML-based classification system to improve campus waste diversion

2023–Present **Mentor**, *IoT-SITY REU*, University of California, Irvine
- Managing development of Unity-based disaster simulation platform
- Delivered working prototype integrated with SHIELD digital twin framework

2020–2021 **Teaching Assistant**, *California State University San Marcos*
- Math 422: Introduction to Number Theory (F20)
- CS 231: Assembly Language and Digital Circuits (W21)

Skills

Programming Languages	Python (PyTorch, TensorFlow), C++ (CUDA), Java, JavaScript, SQL
Technologies	Computer Vision, Distributed Systems, Edge Computing, Machine Learning
Tools	Git, Docker, AWS, Google Cloud, Unity, Linux, OpenCV
Soft Skills	Technical Leadership, Project Management